Boiler Water Chemistry Review: Findings Summary

To: Plant Operations Manager / Water-Treatment Engineer | Period: 30 Apr – 21 Oct 2026 | Basis: 218 shift samples, 11 parameters, compared against plant specifications

43.7%	4 of 11	4 of 11	54
of all readings out of spec (1,009 of 2,310)	parameters breach on more than 80% of samples	parameters breach on 1.5% of samples or fewer	samples (24.8%) with 6+ simultaneous breaches

Bottom line. Core boiler-water control (phosphate, TDS, silica) was largely within specification. Out-of-spec conditions concentrate in upper-drum chemistry, feed-water pH, dissolved oxygen and feed-water temperature, plus a boiler-water pH dip in October. Day and night breach rates are nearly identical for most parameters, which points to system, treatment or specification causes rather than shift practice, with one exception: feed-water pH.

#	Finding	Evidence from the data	Recommended action
1	**Upper-drum pH and TDS are persistently off specification**	bud_ph breached 87.6% of samples, always below the 7.0 limit (mean 5.88). bud_tds breached 53.2%, always above the 2.0 limit (maximum 9.0).	Review treatment and blow-down frequency for the upper drum. This is the most persistent genuine chemistry deviation.
2	**Boiler-water pH dipped in October**	bw_ph breached on 19 of 38 October samples (50%) against 10% or less in May to September. October mean 8.73 against 9.3 to 9.8 earlier; weekly mean recovered to 9.04 by the last week.	Check what changed in October (dosing, make-up water, load) and confirm the recovery holds.
3	**Feed-water pH differs by shift**	fw_ph breached 52.7% on day shift against 22.2% on night shift. Of 82 breaches, 75 were below 8.0.	Compare dosing, sampling time and handover practice between shifts.
4	**Dissolved oxygen behavior changed on 27 May**	Both DO readings were exactly 0.0 until 26 May, then non-zero on almost every sample (no zero readings June to September). Breach rates: bw_do 83.5%, fwt_do 84.4%. The specification is 0.	Check analyzer calibration or replacement records and deaerator performance for late May. Confirm the specification against the instrument's detection limit.
5	**Feed-water temperature breaches every sample**	217 of 218 readings were above 100 °C; 76.6% sat between 100 and 102 °C (median 100.7 °C); 18 exceeded 105 °C. One reading of 73.2 °C (31 Aug) followed a shutdown.	Confirm the 90 to 100 °C specification against design intent. Investigate the 18 readings above 105 °C.
6	**Boiler-water control is strong, with data caveats**	phosphate 0.9% breach, but 213 of 218 readings are recorded as 6.7. bw_tds 0.5%: one spike (311 ppm, 1 Jun); next highest 250 ppm, 83% of the limit. fw_silica 1.4%, with 78% of readings exactly at the 0.30 limit. bo_silica 1.5%: both breaches were its first two readings.	Maintain current control. Confirm how phosphate is logged. Keep a TDS blow-down trigger below 300 ppm.

How to read this. A breach means a reading outside the specification; it does not measure severity or prove equipment damage. Rates use non-blank readings (bo_silica was first measured on 27 Jun, 130 of 218 samples). Data was transcribed manually from shift logbooks. Concentrations are in ppm, temperature in °C.

This summary is a snapshot as of 21 Oct 2026. The accompanying Excel dashboard is a live monitoring tool: new shift readings added to the source data recalculate every rate and chart on refresh. Full method, dashboard and data: see the project repository README on GitHub; http://www.github.com/joelokolie/boiler-water-chemistry-compliance-and-out-of-spec-diagnostic